Cooperative Control of a Tethered Drone and Surface Vessel for Sustained Presence at Sea

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Abstract

An unmanned surface vehicle and an unmanned aerial vehicle connected by a tether combine the endurance of the vessel with the view from above of the aircraft, which receives power and a reliable communication link through the tether. The tether, however, couples the vehicles, which must keep within its reach while each respects its own limits, under waves and wind. This thesis develops the cooperative control of such a system, formed by a WAM-V vessel and an x500 multirotor. Control-oriented models of both vehicles are derived and validated against a simulator, together with a closed-form model of the force of a slack tether that accounts for its weight. Each vehicle is driven by its own nonlinear model predictive controller, and the two controllers exchange their predicted trajectories, so that each keeps the tether within its reach while anticipating the motion of the other. A disturbance observer compensates unmodelled forces, such as wind. The system is evaluated in a simulation at sea built on Gazebo, PX4 and MoorDyn. The aerial controller tracks a reference with less than a quarter of the error of the position controller of PX4, and both controllers plan, from simple or unreachable references, motions within the limits of the vehicles and of the tether. In missions of scouting, inspection of a tower and pursuit of a target, the two vehicles work as one system, and the aircraft lands on the moving deck a few centimetres from the pad centre, under waves and winds of up to 12 m/s.

Keywords: Tethered UAV, Unmanned Surface Vehicle, Cooperative Control, Nonlinear Model Predictive Control, Maritime Robotics

1. Introduction

Saving lives at sea, protecting marine ecosystems and securing maritime infrastructure all rely on a continuous and local awareness of what happens at sea [2]. Surface vessels provide a long-lasting presence, but they only see as far as the horizon, while unmanned aerial vehicles (UAVs) sense from above but have a short endurance and an unreliable long-range link [4]. An unmanned surface vehicle (USV) and a UAV working as a team can keep an awareness that neither achieves alone, and a tether between them supplies the UAV with power and a reliable communication link from the vessel, which removes the two main limits of maritime UAVs [8]. Such systems already exist in industry [7, 1], but their operation remains largely manual.

The tether, however, turns the two vehicles into a single system. The distance between them must stay within its reach, so the motion of one restricts that of the other, and the two move on very different time scales, the USV slowly, dominated by its inertia and the drag of the water, and the UAV quickly. Wind and waves disturb both, and the UAV must take off from and land on a deck that moves with the waves. In the literature, the two vehicles are often controlled separately, the tether is simplified, or the vessel is assumed to be fixed.

This work addresses the control of a USV, the WAM-V, and a UAV, the x500 multirotor, connected by a tether, so that they carry out maritime missions autonomously as a single system, each within its own limits and both within the reach of the tether, also under waves and wind. Its main contributions are:

- control-oriented models of both vehicles and a closed-form model of the force of a slack tether, validated against a simulator;
- two cooperating nonlinear model predictive controllers (NMPCs), one for each vehicle, which exchange their predicted trajectories to keep the tether within its reach, and plan the motion of both vehicles from simple or unreachable references;
- a simulation of the tethered system at sea, with waves, the PX4 autopilot and a tether whose length is set by a winch;

- manoeuvres and missions for both vehicles, including the takeoff from and the landing on the moving deck, and the inspection of a structure that lies beyond the reach of the tether from any fixed point of the USV;
- an evaluation of the system in tracking and planning, in three missions, and under waves and wind.

Section 2 reviews related work, Section 3 presents the models and the controllers, Section 4 the simulation, the validation of the models and the missions, Section 5 the results, and Section 6 concludes.

2. Related Work

Tether models. In tethered aerial systems, the tether is most often modelled as massless, inextensible and taut, so that it acts along the straight line between its ends, as in the buoy–UAV system of [13] and the cooperative manipulation of [17]. Alternatively, its effect is not modelled and is lumped into a disturbance estimated online and rejected by the controller [20]. Neither describes a slack tether, whose weight bends it into a catenary [11] and turns its force away from the straight line.

Control and cooperation. Predictive control is the common choice for the cooperation of a UAV and a USV, since it can state the coupling between them as constraints over a horizon. In [14], an NMPC drives a UAV to follow and land on a moving USV, but only the UAV is controlled and the interaction is unidirectional. In [21], the USV follows a predefined path with a line-of-sight guidance law, and the UAV keeps a relative configuration with its own NMPC, in a leader–follower structure in which the motion of the USV ignores the UAV. In [17], a single, centralised MPC plans the motion of both vehicles jointly, but imposes one horizon and one sampling time on two vehicles whose dynamics evolve on very different time scales. Either one vehicle plans without regard to the other, or both are planned as one.

This work addresses each of these points. It models the force of a slack tether in closed form, accounting for its weight, and compensates what the model leaves out with a disturbance observer. Each vehicle is driven by its own NMPC, with a horizon and a sampling time suited to its dynamics, and the two controllers exchange their predicted trajectories, so that each plans its own motion online while anticipating that of the other, within the reach of the tether.

3. Modelling and Control

3.1. USV Model

The controller of the USV only needs to predict its motion on the surface, so the WAM-V is described in the horizontal plane, by its position (x, y) and heading ψ in the inertial frame and by its surge, sway and yaw velocities (u, v, r) in its body frame, while heave, roll and pitch are left as disturbances of the waves. The kinematics are $\dot{x} = u\cos\psi - v\sin\psi$, $\dot{y} = u\sin\psi + v\cos\psi$ and $\dot{\psi} = r$, and the dynamics follow from Newton's and Euler's laws in the rotating body frame, with the linear-plus-quadratic hydrodynamic damping of [9],

$$\begin{aligned} \dot{u} &= \tfrac{1}{m}\big[X - (X_u + X^{\pm}_{uu}|u|)u\big] + vr, \\ \dot{v} &= \tfrac{1}{m}\big[Y - (Y_v + Y_{vv}|v|)v\big] - ur, \\ \dot{r} &= \tfrac{1}{I_z}\big[N - (N_r + N_{rr}|r|)r\big], \end{aligned} \tag{1}$$

where the quadratic surge damping $X^{\pm}_{uu}$ differs between forward and backward motion. The forces and moment (X, Y, N) are produced by two azimuth thrusters, with thrusts T_L, T_R and angles α_L, α_R, mounted at $(-x_{arm}, \pm y_{arm})$,

$$\begin{aligned} X &= T_L\cos\alpha_L + T_R\cos\alpha_R, \\ Y &= T_L\sin\alpha_L + T_R\sin\alpha_R, \\ N &= -x_{arm}(T_L\sin\alpha_L + T_R\sin\alpha_R) \\ &\quad - y_{arm}(T_L\cos\alpha_L - T_R\cos\alpha_R). \end{aligned} \tag{2}$$

3.2. UAV Model

The x500 is described by its position $\mathbf{p} = [x, y, z]^\top$ and attitude (ϕ, θ, ψ). Its acceleration results from the thrust T along its body vertical axis $\mathbf{z}^{\mathcal{B}}$, gravity, a linear rotor drag d_r on the horizontal velocity, and the force of the tether,

$$\ddot{\mathbf{p}} = \frac{T}{m}\,\mathbf{z}^{\mathcal{B}} - \begin{bmatrix}0\\0\\g\end{bmatrix} - \frac{d_r}{m}\begin{bmatrix}\dot{x}\\\dot{y}\\0\end{bmatrix} + \frac{1}{m}\mathbf{F}_{\text{tether}}. \tag{3}$$

The UAV is driven through its autopilot, PX4 [16], at the attitude level, so the controller does not act on the attitude directly but on the setpoint of the autopilot. The direction of the thrust follows the commanded one as a first-order system with time constant τ, after a transport delay T_t, and the yaw follows its command with τ_ψ after the same delay,

$$\begin{aligned} \dot{\mathbf{z}}^{\mathcal{B}} &= \tfrac{1}{\tau}\big(\mathbf{R}(\phi_{\text{cmd}}, \theta_{\text{cmd}}, \psi_{\text{cmd}})|_{t-T_t}[0,0,1]^\top - \mathbf{z}^{\mathcal{B}}\big), \\ \dot{\psi} &= \tfrac{1}{\tau_\psi}\big(\psi_{\text{cmd}}(t - T_t) - \psi\big), \end{aligned} \tag{4}$$

while the thrust reaches the rotors after a shorter delay T_T. In the prediction, each delay is represented by a first-order Padé approximation, which adds one state per delayed command and, unlike a Taylor expansion, never lets a command act on the vehicle before its delay has elapsed.

3.3. Tether Force Model

The tether is attached to a winch on the USV, at $\mathbf{p}_v$, which keeps its length L slightly longer than the distance $d = \|\mathbf{r}\|$ between its ends, $\mathbf{r} = \mathbf{p}_v - \mathbf{p}$, with a constant slack ratio $\varepsilon_0 = (L-d)/L$. A tether of linear mass μ then hangs as a catenary, and the tension at the UAV is close to the weight of the tether, $T_{\text{tether}} \approx \mu g L$, so that

$$T_{\text{tether}} = c_d\, d, \qquad c_d = \frac{\mu g}{1-\varepsilon_0}. \tag{5}$$

The weight of the tether also turns its force away from the chord $\hat{\mathbf{d}}_0 = \mathbf{r}/d$, towards the vertical, by an angle γ. Since only the weight normal to the chord bends the tether, this deflection is largest for a horizontal chord, where it equals the angle γ_0 of the catenary at its ends, and vanishes for a vertical one, so it is taken as $\gamma = \gamma_0 \cos\alpha$, where α is the depression angle of the chord below the horizontal direction $\hat{\mathbf{e}}_h$ in the plane of the catenary. The force on the UAV is then

$$\mathbf{F}_{\text{tether}} = c_d\, d \left[\cos(\alpha+\gamma)\, \hat{\mathbf{e}}_h - \sin(\alpha+\gamma)\, \mathbf{z}^{\mathcal{I}} \right], \tag{6}$$

a closed-form expression of the positions of the two vehicles, smooth for every orientation of the chord, which replaces the numerical solution of the catenary at every step of the controller.

3.4. Cooperative NMPC

Each vehicle is driven by its own NMPC, which solves, at every sampling instant k, an optimal control problem over a horizon of N steps,

$$\begin{aligned} \min_{\hat{\mathbf{x}},\hat{\mathbf{u}},\hat{\boldsymbol{\sigma}}} \; & \sum_{i=0}^{N-1} \ell(\hat{\mathbf{x}}_i,\hat{\mathbf{u}}_i) + \ell_f(\hat{\mathbf{x}}_N) + \sum_i \rho(\hat{\boldsymbol{\sigma}}_i) \\ \text{s.t. } & \hat{\mathbf{x}}_0 = \mathbf{x}_k, \quad \hat{\mathbf{x}}_{i+1} = \boldsymbol{\Phi}(\hat{\mathbf{x}}_i,\hat{\mathbf{u}}_i), \\ & \hat{\mathbf{u}}_i \in \mathcal{U}, \quad h_j(\hat{\mathbf{x}}_i) \le \hat{\sigma}_{j,i}, \quad \hat{\sigma}_{j,i} \ge 0, \end{aligned} \tag{7}$$

where $\boldsymbol{\Phi}$ integrates the model over one sampling interval with a fourth-order Runge–Kutta method. The limits of the inputs are hard, whereas the state constraints are softened by slacks σ_j, penalised by $\rho(\sigma) = \mu_1\sigma + \mu_2\sigma^2$, so that the problem always has a solution, with penalties large enough to keep the violations close to zero whenever the constraints can be met. Both problems are written in CasADi [3] and solved by acados [19].

A reference far from the vehicle, or beyond what its limits or the tether allow, must not drive it against its constraints. The position error therefore enters both costs through a pseudo-Huber function, $\mathbf{e}\,(1+\|\mathbf{e}\|^2/\delta^2)^{-1/4}$ as a residual, which is quadratic near the reference and grows only linearly beyond a distance δ. A distant reference thus pulls the vehicle with a bounded force, which never outweighs the penalty of a constraint, and the vehicle approaches it as closely as its limits allow.

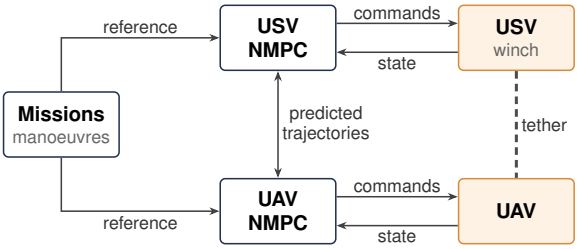


Figure 1: Overview of the system.

A single controller for both vehicles would impose one horizon and one sampling time on vehicles that move on very different time scales, and enlarge the problem to be solved in real time. Each vehicle therefore has its own NMPC, and the two exchange the trajectories they predict, as shown in Figure 1. Each controller then constrains the distance between the vehicles along the predicted trajectory of the other, so that it anticipates where the other vehicle is going, rather than where it is.

USV NMPC. To limit both the thrusts and their rates, the thrusts and angles are included in the state, $\mathbf{x}_v = [x, y, \psi, u, v, r, T_L, T_R, \alpha_L, \alpha_R]^\top$, and the inputs are their rates, bounded by the limits of the thrusters. The stage cost penalises the errors of position, heading and velocity, together with the thrusts and their rates. A soft lower bound on the surge speed limits the motion astern, for which the hull and the thrusters are least effective, and the separation from the predicted trajectory of the UAV, $\hat{\mathbf{p}}_a$, is limited by the reach of the tether,

$$(x-\hat{x}_a)^2 + (y-\hat{y}_a)^2 + (\hat{z}_a - z_t)^2 - d_{max}^2 \le \sigma, \tag{8}$$

where z_t is the height of the winch.

UAV NMPC. The commanded attitude and thrust are included in the state, together with the states of the delays, and the inputs are their rates. The model combines (3)–(6), with the tether attached to the predicted trajectory of the USV. The stage cost penalises the errors of position and velocity, the tilt, the deviation of the thrust from hover and the inputs, and keeps the UAV looking at the winch. The soft constraints bound the height, the horizontal and vertical speeds and the tilt from the vertical, $\cos\phi\cos\theta \ge \cos\phi_{max}$, which is imposed on both the attitude and its command, so that the setpoint sent to the autopilot also respects it. The tether adds two more. The separation is limited, $d \le d_{max}$, and so is the rate at which it grows, $\dot{d} \le \dot{d}_{max}$, which may not exceed the pay-out speed of the winch.

Disturbance observer. A force that the model does not describe, such as wind, makes the predic-

tion drift and leaves a steady offset. Following the disturbance-model approach to offset-free predictive control [18, 15], an external force $\hat{\mathbf{F}}_{\text{ext}}$, held constant over the horizon, is added to (3) and estimated from the one-step prediction error of the velocity,

$$\hat{\mathbf{F}}_{\text{ext},k} = \text{sat}_{\bar{F}}\Big(\hat{\mathbf{F}}_{\text{ext},k-1} + \beta\frac{m}{T_s}\big(\mathbf{v}_k - \hat{\mathbf{v}}_{k|k-1}\big)\Big), \quad (9)$$

where the gain β sets how quickly the estimate follows the force, with a time constant of about T_s/β, and the saturation $\bar{F}$ keeps it within the forces the UAV can expect.

4. Implementation

4.1. Simulation Environment

The system is simulated in Gazebo [12], with the hydrodynamics and the waves of the Virtual RobotX environment [5], in which the WAM-V heaves, rolls and pitches on the waves. The x500 is flown by PX4 in software-in-the-loop, so that the NMPC drives it through the same autopilot as on the real vehicle. The tether is simulated with MoorDyn [10] as a chain of lumped masses with its own weight, elasticity and drag, so that it pulls on both vehicles through its own dynamics rather than through the model of the controllers. A virtual winch on the deck pays it out or reels it in at up to 2 m/s, and holds the UAV against the deck during the takeoff and the landing. Wind acts on the UAV as a drag force with gusts.

4.2. Model Validation

The models of Section 3 were compared with the simulator as the controllers use them, over predictions as long as each horizon, started from the state of the simulator and driven by the recorded commands, as listed in Table 1. Over its horizon, the model of the WAM-V predicts its position within a few centimetres, and that of the x500 within about one percent of the distance it travels, an accuracy that relies on the delays of the attitude response. The model of the tether reduces the error of the force by about three quarters with respect to a taut tether or a hanging load, and follows the tension computed by the simulator from the dynamics of the tether using only the positions of the two vehicles, as shown in Figure 2.

4.3. Controller Implementation

The horizons follow the dynamics of each vehicle. The USV NMPC runs at 10 Hz over 2 s, long enough to see the response of a slow vessel to its thrusters, and the UAV NMPC at 50 Hz over 1 s. The attitude of the UAV starts to respond to a command only 0.12 s after it is sent, longer than the response

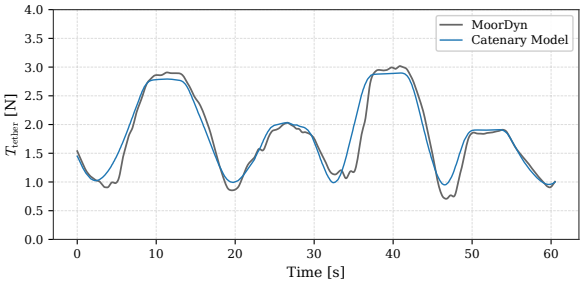


Figure 2: Tension of the tether at the UAV, computed by MoorDyn in the simulator and by the model of Eqs. (5)–(6).

Table 1: Errors of the models against the simulator (RMSE, and NRMSE relative to the range of each signal).

Model	**Quantity**	**RMSE**	**NRMSE**
WAM-V, 2 s	position	1.4 cm	0.37 %
	heading	0.16°	0.33 %
x500, 1 s	position	10.8 cm	1.34 %
	velocity	0.17 m/s	2.02 %
	roll, pitch	0.63°, 0.40°	1.32, 0.89 %
Tether force	model	0.25 N	10.8 %
	taut tether	0.90 N	38.9 %
	hanging load	1.03 N	44.7 %

itself takes, which is why the delays are part of the prediction.

Each controller resamples the latest predicted trajectory of the other at the nodes of its own horizon, since the two run at different rates. The horizon of the UAV covers only the first second of that of the USV, so the USV enforces the tether constraint only within that window, which slides forward as the problem is solved again at every instant. The USV solver iterates to convergence within its sampling interval. The UAV solver could not always do so within 20 ms, so it performs a single iteration per instant, the real-time iteration [6], whose commands stay close to the converged ones. If a solver fails, it is re-initialised at the current state and solved once more.

4.4. Manoeuvres and Missions

The controllers follow references, but do not decide where each vehicle should go. This is the role of the manoeuvres, which give the reference of one vehicle over the horizon of its controller at every instant. The UAV can take off, fly in formation with the USV, orbit it or a point in the world, chase a target and land, while the USV can hold its position, follow a route, pursue a target, or support the UAV from the water near a structure, so that the structure stays within the reach of the tether. Since the controllers keep the vehicles within their limits, the references can remain simple, often a single point. Figure 3 shows the manoeuvres of the

UAV as flown in the missions, the takeoff and the landing through the height of the UAV above the winch, and the others from above.

The takeoff and the landing are the manoeuvres in which the UAV is closest to the deck and moves with it. The takeoff starts with a vertical climb while the winch holds the UAV against the deck, so that it does not drift over its edge at low height. In the landing, the UAV first waits above the pad until it is centred and moving with it, then descends with the winch pulling it, and ends with a slow approach towards a point below the deck, so that it touches down even if the deck falls away.

A mission is a file with a sequence of manoeuvres for each vehicle, in which each manoeuvre ends with an event, such as the detection of a target, or with the end of a manoeuvre of the other vehicle. Three missions were tested: a scout, in which the UAV flies ahead of the bow while the USV follows a route, an inspection, in which the UAV circles a tower while the USV supports it, and a patrol, in which the UAV orbits the USV until it detects a target, which it then chases while the USV pursues it.

5. Results

All results were obtained in simulation, with the default sea of the simulator, which moves the winch up and down by about 0.25 m, without wind unless stated otherwise, and with the state of both vehicles taken from the simulator.

5.1. Tracking

Each controller was first evaluated on a reference within the limits of its vehicle. The USV sails a figure-eight at the speed of the missions, with the UAV on its deck, and follows it within 5.6 cm. The UAV flies a closed trajectory around the USV, which holds its position, made of two fast arcs at constant distance joined by two slow turns, so that its speed and the force of the tether vary along each lap. As a baseline, the same trajectory was flown by the position controller of PX4, which acts on the error only once it has appeared. The NMPC tracks the trajectory with less than a quarter of the error of PX4, as listed in Table 2. The error of PX4 grows with the speed of the reference, which indicates a time delay that the prediction of the NMPC avoids, as shown in Figure 4.

Table 2: Position error of the UAV (cm).

Controller	**RMSE**	**Max.**	**Horizontal**	**Vertical**
NMPC	6.5	12.8	6.4	1.0
PX4	29.9	53.8	29.9	1.8

A second reference for each vehicle demands more than it can deliver. The UAV is given a tight arc that requires more tilt than allowed, a fast arc beyond its speed limit, and two steps in height, the second of which ends below its minimum height, as shown in Figure 5. On each arc, the NMPC brings the UAV onto the limit that makes the reference unattainable and holds it there without oscillation, at the cost of a lag that it recovers once the reference stops. On the way down, it foresees the minimum height within its horizon and decelerates in advance, so that the UAV comes to rest at 2 m although the reference lies on the water.

The USV is given a fast leg, a sharp turn and two steps, astern and abeam, which drive both thrusters to their limits of thrust and angle, as shown in Figure 6. In both vehicles, the limits of the inputs are never exceeded, and those of the states by less than 1 %, so that each controller uses the full capability of its vehicle without driving it beyond its limits.

5.2. Planning

To test the planning, both vehicles were given step references, single points held fixed over the horizon, with $d_{max} = 15$ m. The UAV is sent 10 m sideways, then 12 m up, to a point 22 m from the winch, beyond the reach of the tether; the USV is then sent 18 m away, and both finally return, while the separation stays within its limits, as shown in Figure 7.

When its reference lies beyond the reach of the tether, the UAV stops short of it, on the line towards it, at 14 m from the winch, where the projection of its reference places it, and when the USV then sails away, it follows the USV, with the separation between 14.0 and 14.4 m. In the first step, moving sideways at the same height would separate the vehicles faster than the winch pays out the tether, and the UAV drops by 2 m on the way to slow the growth of the separation, whose rate exceeds its limit of 2 m/s only briefly at the start of the step. The steps bring the actuators to their limits: the thrust of the UAV saturates at its maximum and its commanded tilt at 28.6°, while the thrusters of the USV reach their maximum reverse thrust and their maximum angle as it turns in place. The rates of all the commands reach their limits without exceeding them, and the real pitch of the UAV exceeds its limit by at most 1.7°, as the attitude loop of the autopilot slightly overshoots its command. The controllers therefore do not need feasible, or even continuous, references, which keeps the manoeuvres simple.

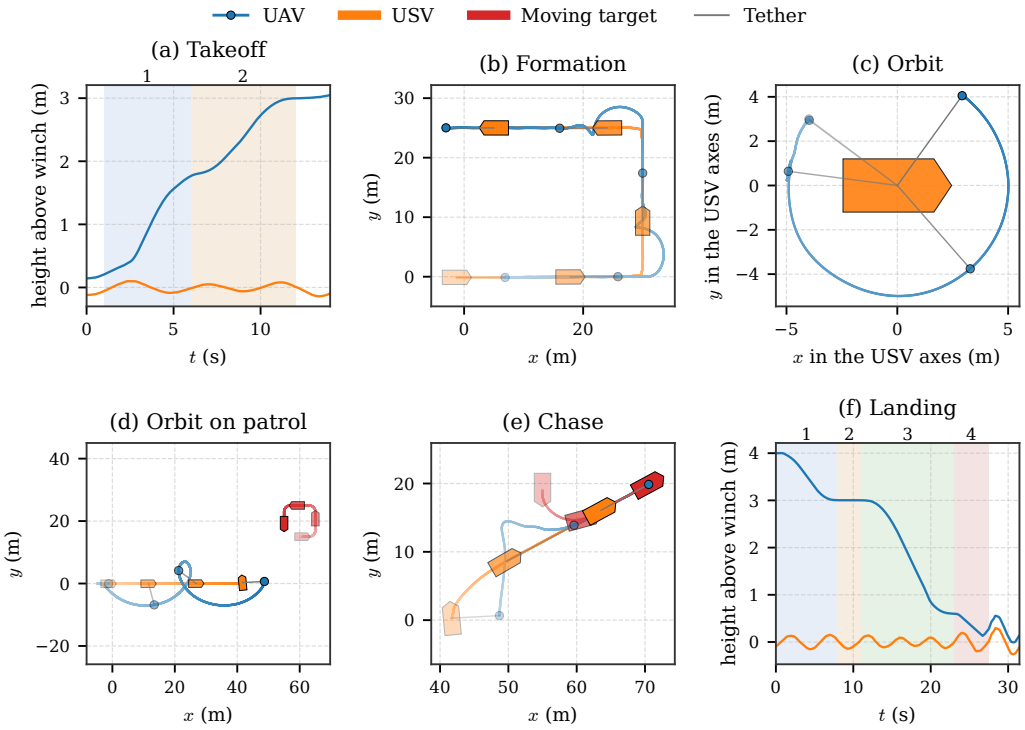


Figure 3: Manoeuvres of the UAV in simulation.

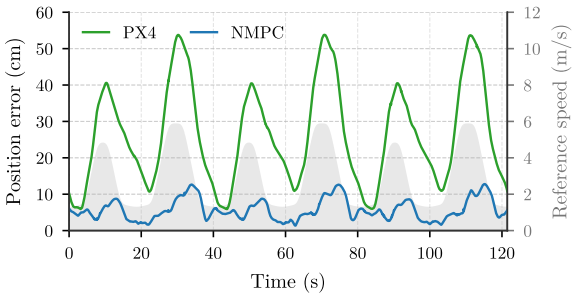


Figure 4: Position error of the UAV over three laps, with the speed of the reference shaded.

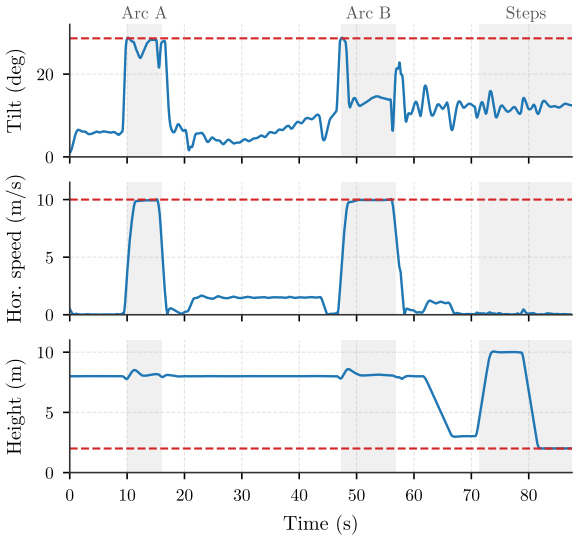


Figure 5: Limits test of the UAV.

5.3. Missions

The three missions were flown with $d_{max} = 15\,\text{m}$, and are summarised in Table 3. In the scout, the UAV keeps 8 m ahead of the bow of the USV as it sails a route with two turns of 90°, with a mean error of 5 cm, largest at the turns, where the point ahead of the bow swings around the USV. In the patrol, the UAV orbits the USV until it detects a target fleeing at 1.5 m/s, 19 m away from it and 23 m from the USV, beyond the reach of the tether. Its reference above the target is brought within reach along the line from the USV, which also lowers it, so that the UAV drops from 10 to 5 m as it flies towards the target, and is above it 5 s after the detection, with the separation at 14.7 m. As the USV closes in, at up to 2.2 m/s, the reference comes back within reach and the UAV climbs to its height again, and the USV catches up with the target 21 s after the detection.

The inspection shows best how the two vehicles work as one system, as shown in Figure 8. The UAV circles a tower of 12 by 13 m at a radius of 13 m, a circle beyond the reach of the tether from any fixed point. Its reference is a circle fixed in the world, regardless of the USV, while the USV supports it from the water, 6 m further from the tower. The USV keeps within 3 cm of its intended radius around the whole tower, the separation stays

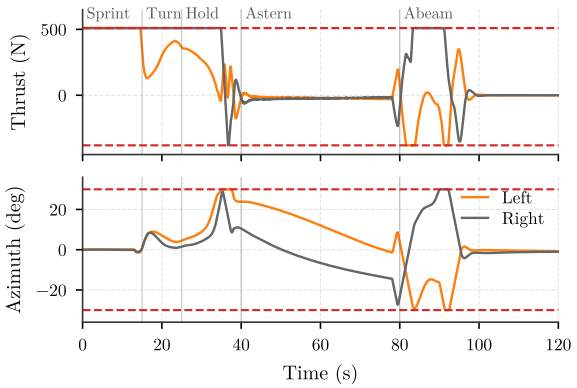


Figure 6: Limits test of the USV.

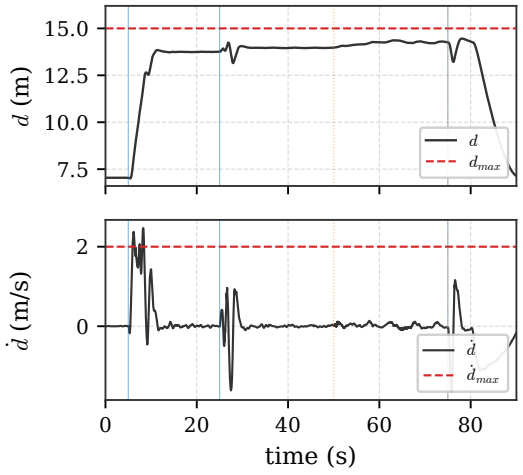


Figure 7: Separation of the vehicles and its rate in the step test.

Table 3: Missions with $d_{max} = 15$ m: duration, largest separation, and distance of the touchdown from the centre of the pad.

Mission	Duration	Max. d	Touchdown
Scout	132 s	9.6 m	3 cm
Inspection	159 s	12.5 m	4 cm
Patrol	104 s	14.7 m	5 cm

at 12.3 m, and the UAV follows its circle within 1 cm, so that the two vehicles together cover a circle that neither could cover alone. In the inspection and the patrol, the roll of the UAV reaches 32° and 34° for less than 0.5 s, above its limit of 28.6°, when its yaw lags behind its command, by up to 13° in the patrol: the tilt is limited about axes turned by the commanded yaw, while the total tilt stays below the commanded one.

5.4. Waves and Wind

The disturbances were added during the landing on the deck of the USV, sailing a route at 0.75 m/s. With the gain of the wave field set to 0, a calm sea, to 1, the default, and to 2, the winch rises

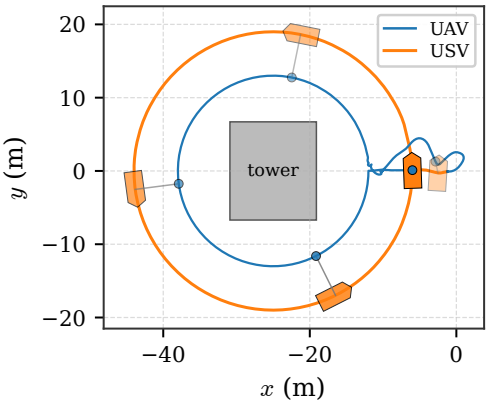


Figure 8: Inspection of a tower, with the vehicles and the tether at five instants.

and falls by 0, 0.25 and 0.44 m, as listed in Table 4. The landing takes the same 25 to 27 s in all sea states, since the reference is taken above the mean height of the winch, and the UAV touches down 0, 2 and 4 cm from the centre of the pad. What the waves change is the contact: the final approach aims below the deck, so the speed of the contact is set by its motion, and grows from 0.15 to 0.26 and 0.39 m/s.

The wind was applied as a drag force on the UAV, with means of 0.9, 2.4 and 5.3 N for 5, 8 and 12 m/s and gusts of ±30 %, with and without the disturbance observer, as shown in Figure 9. Without the observer, the UAV settles away from its reference, 18, 44 and 83 cm from the centre of the pad before the descent, the landing waits for a centring that never happens, so that at 5 and 8 m/s it takes more than twice as long and ends 7 cm off the centre, and at 12 m/s the mission failed in both runs, once at the takeoff and once when the UAV, landing 0.8 m off the centre, was blown off the edge of the pad. With the observer, whose estimate follows the force of the wind about 0.7 s behind it, gusts included, the offset is 4, 6 and 19 cm and the UAV lands within 8 cm of the centre. The most exposed moment is then the takeoff, when a strong crosswind can tip the UAV against the pad while the winch still holds it, before the controller has any effect; the NMPC recovers it once it is released, after a drift of 5.5 m.

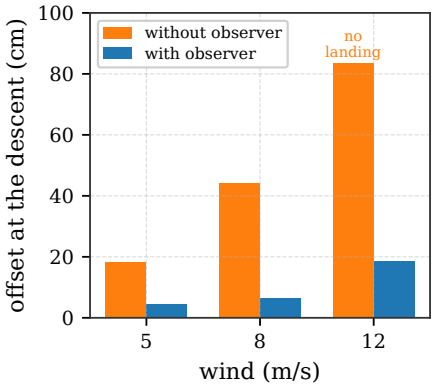


Figure 9: Offset of the UAV from the winch when the descent starts, with and without the observer.

Table 4: Landing under waves and wind: rise and fall of the winch, horizontal offset before the descent, distance of the touchdown from the centre of the pad, and speed of the contact. Without and with the observer, separated by a slash.

Waves (gain)	Heave	Touchdown	Contact
0	0 m	0 cm	0.15 m/s
1	0.25 m	2 cm	0.26 m/s
2	0.44 m	4 cm	0.39 m/s
Wind	**Force**	**Offset (cm)**	**Touchd. (cm)**
5 m/s	0.9 N	18 / 4	7 / 1
8 m/s	2.4 N	44 / 6	7 / 2
12 m/s	5.3 N	83 / 19	failed / 8

6. Conclusions

This work developed the cooperative control of a USV and a UAV connected by a tether. Models of both vehicles, validated against the simulator within 1 % and 2.5 % of the range of their states over the horizons of the controllers, and a closed-form model of the force of a slack tether, more than 70 % more accurate than a taut tether or a hanging load, are used by two NMPCs, one for each vehicle, which exchange their predicted trajectories to keep the tether within its reach. The results show that the NMPC of the UAV tracks a reference with less than a quarter of the error of the position controller of PX4, that both controllers plan, from single points or unreachable references, motions within the limits of both vehicles and of the tether, and that the two vehicles carry out missions as one system, as when the USV supports the UAV around a tower that the tether alone could not reach. The UAV lands on the moving deck within a few centimetres of the centre of the pad under waves of up to 0.44 m, and, with the disturbance observer, under winds of up to 12 m/s, which without it prevent the mission.

The system was evaluated in simulation only, with the state of the vehicles taken from the simulator, and the next step is to test it with the real vehicles. The results also point to the takeoff in strong wind as the most exposed moment, to be addressed by the way the UAV leaves the deck rather than by its controller, and to obstacles, which could be added to both NMPCs as further constraints, on the vehicles and on the tether. Finally, the missions are fixed sequences of manoeuvres, and missions that branch, for instance when a target is lost, would make the system ready for real operations.

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